

Unit 12, Lesson 2: Multiplying Monomials by Algebraic Expressions

Benchmarks

MA.B.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

MA.B.AR.1.2 Apply properties of operations to multiply two linear expressions with rational coefficients.

Learning Targets

- ☐ I can apply properties of operations to multiply two linear expressions.
- ☐ I can multiply a monomial by an algebraic expression.

12.2.1 Warm-Up: Which One Doesn't Belong?

1. Julius wrote a list of expressions using variables, numbers, and operators.

a. Circle the expression that does not belong.

$$h(4 - h)$$

$$(c + 1)(-c)$$

$$2(-3 - x)$$

$$-5(2 \cdot \pi)$$

b. Explain why the expression does not belong.

12.2.2 Collaboration: Applying Properties of Operations to Multiply Linear Expressions

1. With your partner, determine who is Partner A and who is Partner B.
 - a. Apply the distributive property to re-write each expression in your column in the space provided below each given expression.

Partner A
$4r(3 - 8r)$
$-\frac{2}{3}x(6x + 9)$
$-3c(5 - c)$
$(2b - a)12a$
$\frac{1}{4}x(10 + 16x)$

Partner B
$2r(-16r + 6)$
$(-2x - 3)2x$
$c(3c - 15)$
$-6a(2a - 4b)$
$-0.5z(-8z - 5)$

- b. Compare your work with your partner. The expressions in each row are equivalent. If your expressions are not equivalent, work together to find and correct any error(s).

12.2.3 Collaboration: Defining Monomials

1. The charts below show several examples and non-examples of monomials.

Monomials

r^5 5 x
 $\frac{3}{4}x^2y$ $80z^{15}$

Not Monomials

$2.7r^{-3}$ $3x + 2y$
 $\frac{x^2}{5y}$ $a^2 + b^2 + 10$

a. Create a list of identifying features monomials can have versus features they cannot have.

Monomials Can Have ...	Monomials Cannot Have ...

b. Compare your lists with your partner. Add any additional features you found after your discussion.

c. Work with your partner to write a definition of a monomial.

- d. Ask two other pairs of partners about the definition they wrote. Listen to their definition and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

My Summary of the Definition of a Monomial	Initials

- e. Review the definitions with your partner and make any necessary adjustments to your own definition.

2. Work with your partner to complete the statement. Select all that apply.

A monomial is a

- ☐ single number
- ☐ single variable
- ☐ sum of numbers and variables
- ☐ product of numbers and variables

with

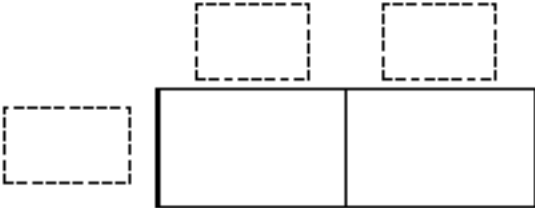
- ☐ whole number
- ☐ integer
- ☐ rational

exponents.

12.2.4 Guided Instruction: Multiplying a Monomial by a Linear Expression

Monomials can also be multiplied by linear expressions by applying properties of operations and laws of exponents.

1. Consider the monomial $-\frac{2}{5}m^2k$ and the linear expression $(4m - 7)$.
- a. Determine the product of $-\frac{2}{5}m^2k$ and $(4m - 7)$ using both methods in the table.

Method 1: Use an Area Model	Method 2: Apply the Distributive Property
	$-\frac{2}{5}m^2k(4m - 7)$

- b. Write another expression equivalent to the product.

12.2.5 Guided Practice: Multiplying a Monomial by a Linear Expression

Find each product using your preferred method.

1. $-2.4c^5(-9c - 3.16)$

2. $\frac{3}{8}xy^4\left(\frac{1}{2}y - \frac{3}{2}\right)$

3. $4.79p^3q(0.3 + 2.1r)$

4. $(5a - 7.8)\left(-\frac{5}{2}a^4b^2\right)$

12.2.6 Your Turn!

Find each product using your preferred method.

1. $-4y^4z\left(\frac{a}{2} - 8z\right)$

2. $(12.7f + 9.2)(4.01f^2g^3)$

3. $\frac{4}{5}m^4n^2\left(\frac{7}{2}m - 10\right)$

4. $(2.005 - x) \cdot (x^5yz^3)$

12.2.7 Wrap-Up: Error Analysis

1. Elijah multiplied the monomial $-\frac{3}{4}x^2y$ and the linear expression $\frac{2}{5}x - \frac{8}{2}$. His resulting expression is shown.

$$-\frac{3}{10}x^3y - \frac{8}{2}$$

Write a note to Elijah explaining what he did correctly and what error(s) he made when finding the product of $-\frac{3}{4}x^2y$ and $\left(\frac{2}{5}x - \frac{8}{2}\right)$.

